

Muhammed Shanoob A K

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Profile Summary

Detail-oriented and self-taught Full-Stack Engineer with 3.5 years of professional technology experience including web development, cloud infrastructure, and system design. Proficient in TypeScript, Node.js, Express, React, PostgreSQL, and MongoDB, with expertise in implementing clean architecture principles, designing relational data models, and building modular systems with asynchronous communication using RabbitMQ, Redis, and Socket.IO for real-time features. Experienced in Microsoft Graph API integration for enterprise meeting management—including webhook-driven event processing, OAuth flows, and asset streaming. Proven ability to deliver end-to-end solutions—from UI components to backend APIs—ensuring performance, reliability, and maintainability. Passionate about solving complex technical challenges and building systems that are efficient, secure, and user-friendly.

Technical Skills

Backend: Node.js, Express.js, TypeScript, PostGraphile, Awilix (DI), REST APIs, Clean/DDD Architecture

Database: PostgreSQL, Prisma ORM, MongoDB

Frontend: React, TypeScript, JavaScript (ES6+), Redux, Redux Toolkit, Material UI (MUI v7), HTML5, CSS3, Vite, Tailwind CSS, Bootstrap

Integrations & Tools: Microsoft Graph API, MSAL, RabbitMQ, Redis, Socket.IO, Server-Sent Events (SSE), JWT, Zod, Nodemailer, Cloudinary

DevOps & Deployment: Docker, GitHub Actions CI/CD, AWS (EC2, ELB, Route53, Secrets Manager), NGINX, Cloudflare, ngrok

Version Control & Tools: Git, GitHub, Postman, Figma, DBDiagram

Key Concepts: Authentication & Authorization (JWT + RBAC + API Keys), Message Queues (DLQ/Retry), Event-driven Architecture, Webhook Systems, Clean Code, Atomic Design

Professional Experience

Software Engineer, Packapeer Academy Pvt. Ltd. – Bangalore, India Oct 2024 – Present

- Contributing as a full-stack engineer in the development of scalable, enterprise-grade web applications using TypeScript, React, Node.js, and PostgreSQL under clean and DDD architecture principles
- Designed and implemented PostgreSQL database schemas, complex functions, and views using joins, CTEs, and advanced SQL techniques
- Developed frontend and backend modules using React + Material UI (Atomic Design) and Node.js/Express/PostGraphile APIs with type safety and modularity
- Integrated RabbitMQ for inter-service communication — implemented producers, consumers, retry/DLQ mechanisms, and a centralized email system using Nodemailer for asynchronous, scalable communication
- Implemented an in-app notification system using Redis, Socket.IO, and Server-Sent Events (SSE) for real-time updates
- Built key features including attendance and leave management systems, internal ticketing and payment tracking modules, and role-based access control (RBAC) with refresh token-based authentication
- Architected and built a production Microsoft Teams meeting management microservice using Clean Architecture with strict dependency inversion, Awilix DI container, repository pattern, and engine layer for complex domain orchestration
- Integrated Microsoft Graph API via MSAL client-credential OAuth for server-to-server meeting lifecycle management — create, update, cancel meetings with organizer resolution and service account fallback
- Designed a webhook subscription system for tenant-wide callRecord, recording, and transcript event processing with automatic renewal cron, watchdog reconciliation, and orphan cleanup
- Implemented a RabbitMQ topic exchange architecture with per-command queue triplets (main → retry → DLQ), 5-phase message processing pipeline with Zod validation, Redis-based idempotency, and exponential backoff retry
- Built multi-layer authentication — JWT (HS256) for users, API key for business apps, clientState for Microsoft webhooks — with DB-driven RBAC, permission version sync, three-tier rate limiting, and nonce-based CSRF protection

- Built an asset streaming proxy for Graph API recordings/transcripts with HTTP Range request support, cross-domain redirect handling, and auto-refresh of stale content URLs
- Implemented comprehensive error handling with custom error hierarchy, Graph API retry strategy (401 token refresh, 429 Retry-After, 5xx exponential backoff), error sanitization middleware, and persistent SyncError tracking

TechOps Support Engineer, Uroline Technologies Pvt. Ltd. – Kochi, India

2021 – 2023

- Administered Linux-based environments, optimized server performance, and collaborated with developers on deployment pipelines
- Managed AWS and GCP cloud infrastructure, Kubernetes clusters, and load balancing using NGINX and ELB
- Worked with Git, GitHub, and CI/CD workflows for deployment automation
- Ensured secure and stable hosting for Node.js and web applications
- Hands-on with Cloudflare, Route 53, and SSL configurations for optimized content delivery

Projects

Teams Meet Manager – Enterprise Meeting Microservice

- A production microservice for managing Microsoft Teams meetings end-to-end — lifecycle management, webhook-driven event processing, recording/transcript sync, and chat archiving via Graph API
- Clean Architecture backend (Node.js/TypeScript) with Awilix DI, PostgreSQL/Prisma (24 normalized tables with full audit trail), RabbitMQ command bus with DLQ/retry, Redis caching, and Docker containerized deployment
- React/MUI frontend with permission-gated UI, code splitting with lazy-load auto-recovery, dual light/dark theming, and DB-driven route protection
- Tools Used: Node.js, TypeScript, Express, PostgreSQL, Prisma, React, Redux Toolkit, MUI v7, RabbitMQ, Redis, Microsoft Graph API, MSAL, Docker, GitHub Actions, AWS

Vergno – Academic Social Media Platform

[GitHub](#)

- A social platform for students and professionals to share ideas and engage in topic-based discussions
- Built a real-time chat system with unread message tracking via Socket.IO and Redis
- Implemented global search across posts, topics, and users
- Added Firebase authentication, post management, reporting, and admin moderation tools
- Tools Used: React, Node.js, Express, MongoDB, Firebase, Material-UI, Socket.IO, Redis, Docker, NGINX, JWT

Strides – E-Commerce Application

[GitHub](#)

- Developed a full-featured e-commerce platform with OTP-based authentication, wishlist, and multi-payment options (COD, wallet, Razorpay)
- Created admin dashboards with sales analytics and order tracking
- Tools Used: Node.js, Express, MongoDB, EJS, Bootstrap, Razorpay, Socket.IO, AWS, NGINX

Mini Projects

- User Management System – MERN stack app with authentication and admin controls
- Password Generator – Customizable password creation tool with copy functionality
- Task Manager App – React-based task tracker with local storage persistence

Education

Brototype, Calicut, MERN Stack Development Program

2023 – 2024

College of Engineering Adoor, Kerala, B.Tech in Computer Science and Engineering

2015 – 2019

- CGPA: 6.6/10.0

Additional Information

- Open to remote or on-site roles across India
- Passionate about system design, backend architecture, and performance optimization
- Strong problem-solving mindset and team collaboration skills